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INSIGHTS

OgunBiome Inulin Intervention Insight Report

Analysis of Baxter et al. 2019 — Inulin Arm — SRP128128

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Executive Summary

Inulin supplementation resulted in a significant shift in gut microbiota composition, characterised by a marked enrichment of *Bifidobacterium* and *Anaerostipes* (FDR < 0.05), indicating selective stimulation of saccharolytic and butyrate-producing taxa.

Differential abundance analysis revealed two genera as primary responders to chicory-derived inulin supplementation. *Bifidobacterium* exhibited a 3.5-fold increase in relative abundance, consistent with its established role as a primary degrader of inulin-type fructans. *Anaerostipes* showed a 2.2-fold enrichment, representing the most statistically significant finding (adj. p = 0.0001), reflecting cross-feeding interactions driven by Bifidobacterium-derived acetate.

Community Composition

The following figures show gut microbiota abundance at baseline and during inulin supplementation across all detected genera, and the significant responders at FDR < 0.05.

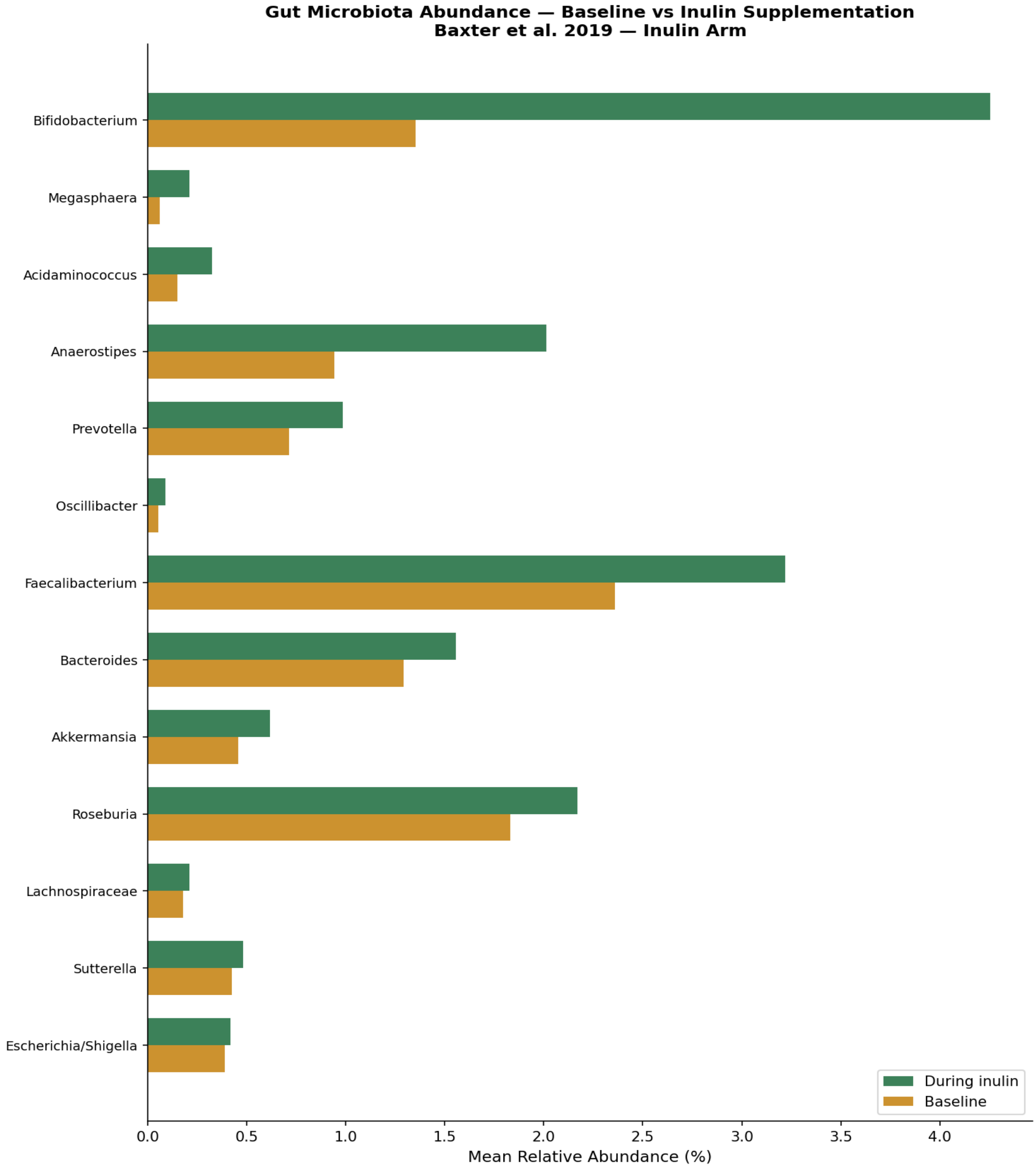

Figure 1. Mean relative abundance of all detected genera at baseline and during inulin supplementation. Sorted by fold change.

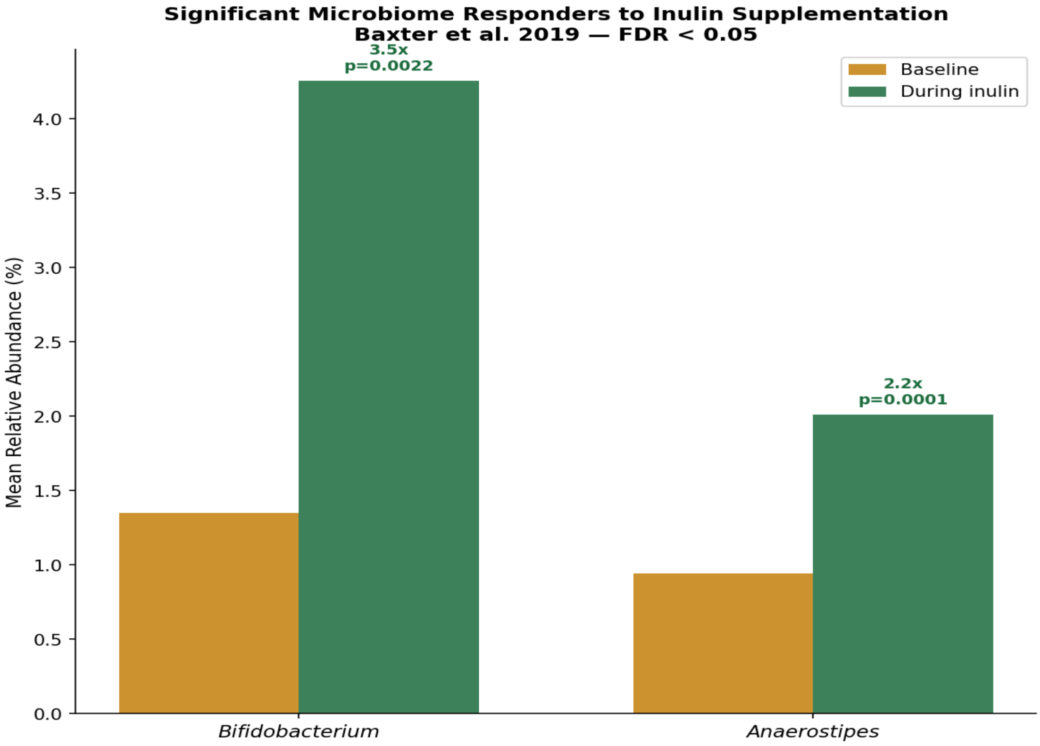

Figure 2. Significant genera — baseline vs during inulin supplementation. Fold change and adjusted p-value annotated.

Differential Abundance Analysis

Volcano plot showing log2 fold change against statistical significance for all 13 genera. Significant genera (FDR < 0.05) highlighted in OgunBiome green and labelled explicitly.

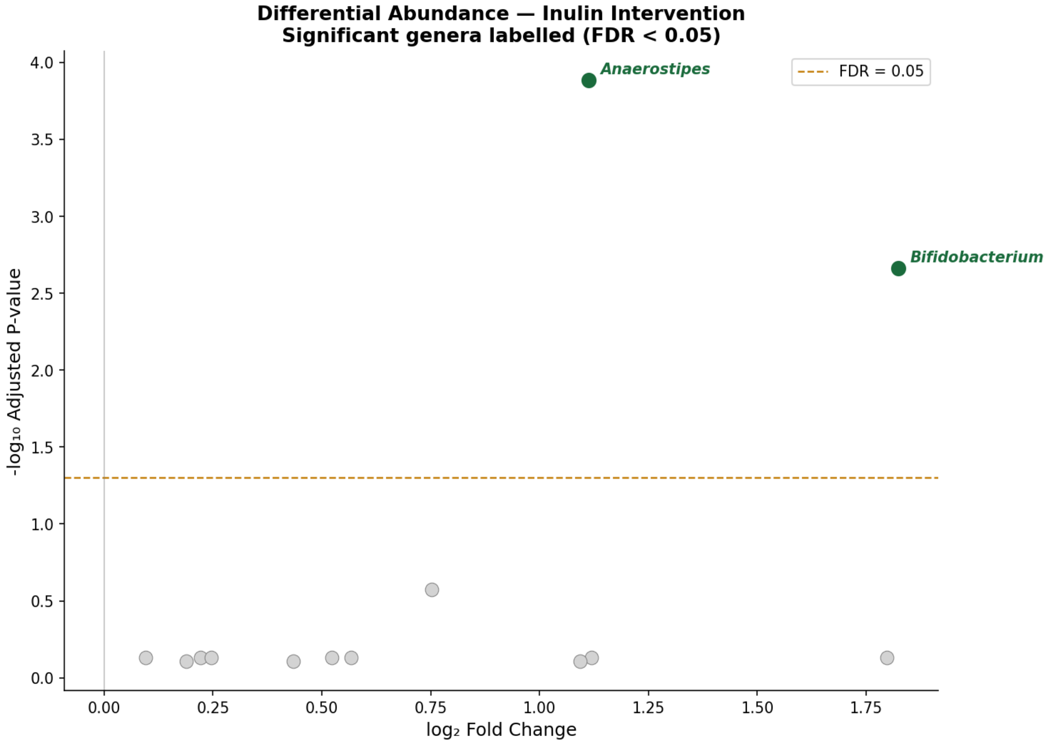

Figure 3. Volcano plot — differential abundance of gut microbiota genera in response to inulin supplementation.

Significant Findings

Genus	Fold Change	Adj. P-value	Direction
Anaerostipes	2.16x	0.0001	Enriched
Bifidobacterium	3.54x	0.0022	Enriched

Biological Interpretation

The following interpretations are provided by Dr. Oluwamayowa Ogun based on domain expertise in gut microbiology, glyco-enzyme biology, and dietary intervention science.

Bifidobacterium

What it is: Bifidobacterium is a gram-positive, anaerobic genus within the phylum Actinobacteria and one of the most well-characterised health-promoting genera of the human gut microbiome.

What changed: Bifidobacterium exhibited a strong increase in relative abundance of approximately 3.5-fold during inulin supplementation (adj. $p = 0.0022$, FDR < 0.05), identifying it as the primary microbial responder to the intervention.

Mechanism: Bifidobacterium species possess beta-fructosidase enzymes that specifically cleave the beta-2-1 glycosidic bonds of inulin-type fructans. This enzymatic capacity positions Bifidobacterium as a primary degrader of chicory-derived inulin, selectively enriched through substrate-specific fermentation activity.

Health relevance: The selective enrichment of Bifidobacterium supports a mechanistic model in which inulin supplementation enhances both primary saccharolytic activity and secondary metabolite production, particularly short-chain fatty acids with established roles in colonic health, gut barrier integrity, and mucosal immune regulation.

EFSA relevance: The enrichment of Bifidobacterium observed here is directly relevant to EFSA scientific opinions on inulin-type fructans and gut health biomarkers. EFSA recognises increased Bifidobacterium abundance as a measurable indicator of prebiotic activity in the human colon, supporting its use as a functional endpoint in dietary intervention studies submitted for health claim substantiation under Regulation EC 1924/2006.

Anaerostipes

What it is: Anaerostipes is an anaerobic, butyrate-producing genus within the family Lachnospiraceae, recognised as a key secondary fermenter in the human gut microbiome.

What changed: Anaerostipes showed a significant enrichment of approximately 2.2-fold during inulin supplementation and represented the most statistically significant finding in the dataset (adj. $p = 0.0001$, FDR < 0.05).

Mechanism: Anaerostipes does not ferment inulin directly. Its enrichment reflects cross-feeding interactions in which acetate and lactate produced by Bifidobacterium during inulin fermentation are utilised by Anaerostipes as substrates for butyrate synthesis via the acetate-to-butyrate pathway. The concurrent enrichment of both genera confirms the activation of this cross-feeding cascade in vivo.

Health relevance: Butyrate produced by Anaerostipes serves as the primary energy source for colonocytes and plays a critical role in maintaining gut barrier integrity, regulating mucosal immune responses, and suppressing pro-inflammatory signalling pathways. The enrichment of Anaerostipes provides biologically plausible support for improved gut metabolic function under inulin supplementation, consistent with established prebiotic mechanisms.

EFSA relevance: The enrichment of butyrate-associated taxa observed here is relevant to EFSA scientific opinions on dietary fibre and gut health, which recognise short-chain fatty acid production — particularly butyrate — as a measurable biomarker of prebiotic activity and a functional indicator of colonic health in human intervention studies.

EFSA Biomarker Mapping

A structured EFSA biomarker mapping layer developed by Dr. Ogun, linking microbiome features to relevant EFSA opinions and guidance documents. Relevance tiers: HIGH — directly cited in positive EFSA opinion; MODERATE — supported by studies cited in EFSA opinions; SUPPORTING — biologically consistent but more distal.

Genus	Tier	EFSA Opinion	Claim Context
Anaerostipes	MODERATE	Dietary fibre and colonic fermentation	Colonic health — short-chain fatty acid production
Bifidobacterium	HIGH	Inulin-type fructans and bowel function	Gut health — prebiotic activity
Bifidobacterium	HIGH	Dietary fibre and gut health	Digestive health — microbiota modulation

Conclusion

This analysis demonstrates that chicory-derived inulin supplementation produces a targeted and biologically coherent restructuring of the gut microbiota, characterised by the selective enrichment of *Bifidobacterium* and *Anaerostipes*. These findings are consistent with the established prebiotic mechanism of inulin-type fructans and are directly relevant to EFSA health claim substantiation under Regulation EC 1924/2006.

The co-enrichment of a primary inulin fermenter and a cross-feeding butyrate producer provides mechanistic evidence of the complete prebiotic metabolic cascade operating in vivo. Butyrate production supports colonic epithelial health, gut barrier integrity, and mucosal immune regulation — outcomes recognised by EFSA as biologically plausible indicators of gut health benefit.

OgunBiome Insights translates microbiome data into actionable gut health intelligence for the food and ingredient industry.